RISHABH SINGH

Data Scientist, Forecasting, Analytics & Applied ML

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SUMMARY

IIT Madras graduate and Data Scientist with 2 years of pharma commercial analytics experience, specializing in time-series forecasting, mixed-effects modeling, and end-to-end ML pipeline delivery. At Chryselys, co-built ForecastIQ, a live commercial pharma forecasting platform, and delivered Streamlit tools adopted by pharma clients. Proficient in Python, SQL, Streamlit, scikit-learn, and Azure. Adept at presenting model outputs to non-technical stakeholders and guiding junior analysts across model validation and production-readiness workflows.

EXPERIENCE

Chryselys
Hyderabad, India

Data Scientist (Forecasting, Analytics & Development)
October 2024 – Present

- **ForecastIQ Product Delivery:** Architected and delivered core product components for ForecastIQ, a live commercial pharma forecasting platform. Owned the dashboard end-to-end and reviewed the scenario planner module.
- **Model Performance and Operationalization:** Designed time-series cross-validation, error-tracking (MAPE, RMSE), and model comparison frameworks.
- **Model Deployment:** Collaborated with backend engineering teams to operationalize statistical models into production-ready pipelines.
- **Streamlit Tools and Enablement:** Engineered a Streamlit-based automation tool that reduced manual preprocessing steps by approximately 40 percent, enabling non-technical stakeholders to independently explore segmentation outputs.
- **FFO Sizing and Mixed-Effects Models (Productization):** Stabilized and extended Mixed-Effects sizing models across 3 or more pharma client engagements. Hardened codebase and authored documentation to enable model reuse in go-to-market sizing exercises.
- **Marketing Mix Modeling (MMM):** Refined feature engineering pipelines and introduced promotion-response features to integrate MMM outputs with ML forecasting models.
- **Cross-Functional Delivery:** Coordinated with Forecasting, Product, and Insights and Analytics teams to ship end-to-end forecasting solutions. Aligned KPIs and streamlined reporting cadence between analytics and commercial teams.
- **Mentorship Cohort:** Coached 6 junior analysts on Python workflows, model validation standards, production-readiness checks, code reviews, and stakeholder-ready analytical presentations.

iNeuron.ai
Remote

Machine Learning Intern
March 2024 – April 2024

- Trained a phishing-domain detection model using Random Forests on approximately 147,000 URLs with full feature engineering, model selection, and validation metrics assessment.
- Packaged the classifier behind a Flask REST API and deployed on Microsoft Azure with a lightweight front-end for real-time URL classification by end-users.
- Authored reproducible evaluation notebooks and workflow documentation to support future extension and handover.

SKILLS

Languages: Python, SQL, R (basics), Bash, JavaScript (familiar)

ML & Statistics: Forecasting (ARIMA, ETS, Prophet, XGBoost), Mixed-Effects Models (LME/LMM), Marketing Mix Modeling (MMM), Regression Analysis, Tree-based Models (Random Forest, Gradient Boosting), SVM, Classification, Clustering, Segmentation, Time-Series Cross-Validation, Feature Engineering, Attribution Analytics, Statistical Modeling, Predictive Modeling, A/B Testing, Hypothesis Testing, Bayesian Inference, NLP, LLM Pipelines, Prompt Engineering

Tools & Frameworks: scikit-learn, pandas, NumPy, PySpark, PyTorch (familiar), TensorFlow (familiar), Streamlit, Flask, FastAPI, Jupyter Notebook, Git, GitHub, Tableau, Power BI (familiar), Figma, Dataiku, Databricks, Celery, Redis, Jira, Bitbucket

Cloud & Infrastructure: Azure (App Service, Functions, Blob Storage), AWS (S3, Lambda, Bedrock, App Runner), PostgreSQL, MongoDB, Docker (familiar), Vercel, Render, REST APIs, CI/CD, MLOps, Data Pipelines, ETL

EDUCATION

Indian Institute of Technology Madras (IIT Madras) B.S. awarded May 2025

B.S. in Data Science and Applications, Minor: Economics and Finance, GPA: 7.89/10

Coursework: Machine Learning, Deep Learning, Neural Networks, Natural Language Processing (NLP), Probability & Statistics, Database Management Systems (DBMS), Software Engineering, Data Structures & Algorithms, Business Analytics, Econometrics, Game Theory, Operations Research

SELECTED PROJECTS

- **LLM YouTube Metadata Connector**, *Python, LLM Pipelines, YouTube API, Claude Desktop, github/LLM-Youtube-Metadata-connecter*
  Engineered an end-to-end LLM pipeline using Claude Desktop and the YouTube Data API v3 to automate metadata generation, SEO-optimized title and description writing, keyword tagging, and content optimization for YouTube; used to update 500+ YouTube videos. Implemented deterministic validation layers, prompt engineering, and a human-in-the-loop review step. Integrated OAuth 2.0 authentication and rule-based post-processing to ensure production-grade reliability of model outputs.
- **Parking Management System**, *Flask, Vue.js, Redis, Celery, Azure, parking.rishabhsingh.me*
  Built a full-stack web application using Flask REST API and Vue.js. Leveraged Redis and Celery for asynchronous task queuing and deployed on Azure App Service with CI/CD. Implemented real-time slot booking, role-based access control, and an operational analytics dashboard.
- **Indian Housing ETL Pipeline**, *PostgreSQL, MongoDB, Python, web scraping*, github/Indian-Housing-ETL-Pipeline
  Designed and implemented a scalable ETL pipeline covering scraping, cleaning, deduplication, and validation. Stored 180,000+ records across PostgreSQL and MongoDB for downstream statistical analysis and machine learning modeling.
- **Mixed-Effects Marketing Modeling**, *Python, statsmodels, MMM, Kaggle*
  Applied hierarchical mixed-effects regression (LME) to quantify marketing campaign elasticity. Decomposed global vs. channel-level effects to support data-driven budget allocation and ROI optimization using core Marketing Mix Modeling (MMM) techniques.
- **Audio Classification (UrbanSound8K)**, *Python, TensorFlow, Librosa, Deep Learning, Kaggle*
  Built an end-to-end deep learning pipeline for urban audio classification. Extracted features using mel-spectrograms and MFCCs via Librosa and trained CNN models with TensorFlow, achieving strong accuracy on the UrbanSound8K benchmark dataset.
- **Compass (Bloglite)**, *Flask, SQLAlchemy, REST API, deployment, compass.rishabhsingh.me*
  Developed a Flask-based multi-user blogging platform with SQLAlchemy ORM, JWT authentication, full CRUD operations, and RESTful API design. Containerized and deployed to cloud with environment-based configuration management.

CERTIFICATIONS & ACHIEVEMENTS

- **Microsoft Certified: Azure Fundamentals (AZ-900)**: cloud computing, Azure services, security, compliance, pricing
- **Microsoft Certified: Azure Data Fundamentals (DP-900)**: relational & non-relational data, Azure data services, analytics workloads
- **Microsoft Certified: Azure AI Fundamentals (AI-900)**: machine learning, computer vision, NLP, conversational AI on Azure
- **HarvardX: CS50 Introduction to Computer Science (2025)**: algorithms, data structures, C, Python, SQL, web development
- **HarvardX: R Basics (2025)**: statistical computing, data wrangling, visualization in R
- **UC Davis: Data Visualization with Tableau (2024)**: dashboards, visual analytics, storytelling with data
- **Rank 3**, IIT Madras CodeChef Programming Competition (2022): competitive programming, problem solving, algorithms
- Open source contributor to **CPython** (python/cpython#134804) and **conda-build** (conda/conda-build#4782)